# DAI-Net: Dual Adaptive Interaction Network for Coordinated Medication Recommendation

Xin Zou, Xiao He, Xiao Zheng, Wei Zhang, Jiajia Chen, and Chang Tang, *Senior Member, IEEE*

***Abstract*—Medication recommendation is a productive task for AI-driven healthcare systems, which can assist clinicians in prescribing judicious and effective treatments. However, existing medication recommendation methods omit two key pieces of information: Coarse-grained interaction information between distinct types of symptoms in a patient's medical history and corresponding medication representations can serve as attention for predicting the current medication combinations of the patient. Fine-grained interaction information between medication substructure representations and different types of symptoms can facilitate the construction of molecular-level disentangled medication representations. To address this dilemma, we propose a novel Dual Adaptive Interaction Network (DAI-Net), which encodes comprehensive interaction knowledge between patients' multifaceted health records and medication molecules to improve the performance of medication recommendation and heighten interpretability of the model. Specifically, we design a symptom-aware medication matching module to extract coordinated associations between patient symptoms and medication molecules, coarse-grained interaction learning. The medication embeddings are utilized to transform patient-medication matching properties into a symptom-substructure matching matrix for fine-grained interaction. The patient's Longitudinal representation is employed as a query to decode both symptom-medication and symptom-substructure matching information for coordinated medication representation. DAI-Net is an end-to-end recommendation model. Extensive experiments on the real-world EHR datasets, i.e., the public benchmark MIMIC-III, MIMIC-IV, and eICU, demonstrate that the proposed DAI-Net achieves competitive performance compared to other state-of-the-art ones, with an average improvement of 1.8%, 2.1% in Jaccard on MIMIC-III and -IV dataset.**

***Index Terms*—Medication recommendation, healthcare, longitudinal EHR, deep learning.**

## I. Introduction

HEALTHCARE sectors have witnessed a remarkable acceleration in digitization, with leading medical institutions accumulating substantial electronic health record (EHR) databases [1] in recent years. The application of artificial intelligence (AI) in analyzing these extensive data repositories holds the potential to yield critical insights, shedding light on the diagnostic and therapeutic approaches of experts at these advanced medical facilities [2], [3], [4]. The extraction of such knowledge could catalyze a plethora of innovative healthcare applications, encompassing sophisticated follow-up consultations [5], [6], medical imaging analysis [7], [8], and management of chronic illnesses [9], [10]. The prospective impact on future healthcare practices is profound.

AI technology's expanding role in diverse medical domains has been instrumental in advancing towards a more equitable and universally accessible healthcare system [11], [12]. Notably, while areas such as medical imaging have realized significant achievements, the applications in medication recommendation systems are comparatively nascent [13], [14]. Despite initial promising developments, numerous challenges and unresolved issues remain, particularly in enhancing the prediction accuracy and interpretability of medication recommendation systems [15], [16]. Furthermore, the need for large-scale, high-quality, real-world datasets for the training and validation of these models is crucial. Additionally, the ethical and privacy considerations in utilizing patient data for model development while safeguarding patient confidentiality are yet to be fully addressed [17], [18], [19]. Overall, despite the preliminary success and prospects, it is still early days for AI in the field of pharmaceutical research.

One specific area of focus is the medication recommendation process using longitudinal multi-source EHR data, which seeks to provide appropriate medication combinations for a series of diagnosed conditions in patients [20], [21], [22]. This approach entails the clinician assessing the patient's medical history over multiple visits and prescribing effective medication

Received 29 November 2023; revised 2 June 2024; accepted 2 July 2024. Date of publication 30 July 2024; date of current version 4 October 2024. This work was supported in part by the National Science Foundation of China under Grant 62076228, in part by the China University of Geosciences (Wuhan) Teaching Laboratory Open Fund, and in part by NVIDIA Corporation with the donation of the Titan V GPU. *(Corresponding authors: Jiajia Chen; Chang Tang.)*

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The implementation is available at: https://github.com/12xzou/DAI-Net.

Digital Object Identifier 10.1109/JBHI.2024.3425833

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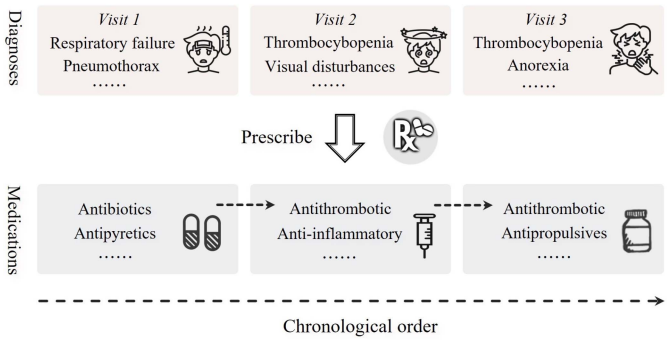



Fig. 1. A toy example of a longitudinal electronic health record that consists of three visits by a patient.

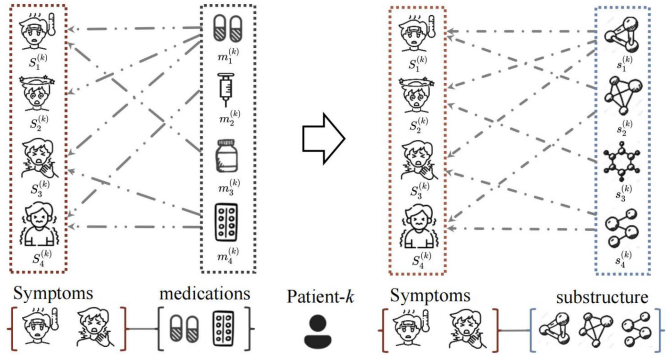



Fig. 2. Schematic illustration of dual interaction proposed in this work. Symptom-medication and symptom-substructure matching matrices are obtained through coarse- and fine-grained interaction learning between patients and medications. After that, the optimum medication combination is located based on the symptom information of the patient-$k$, i.e., the patient's representation is used as a query to calculate the medication representation from the two matching matrices.

combinations tailored to the identified conditions, reflecting a nuanced understanding of patient needs over time. As illustrated in Fig. 1, in this process, clinicians evaluate the evolution and interplay of various health conditions over time, thereby enabling them to prescribe medication combinations that are not only effective but also specifically tailored to the individual patient's evolving health scenario [23].

Medication recommendation [24], [25], different from conventional product recommendation [26], faces two primary challenges: (a) Traditional recommendation systems, typically based on collaborative filtering, are designed to suggest individual items to users by processing single item-user representations and generating a matching score [27], [28]. However, medication recommendation necessitates the simultaneous output of medication combinations, a significantly more complex task. (b) The interactions among drugs are intricate, with potential synergies, antagonism effects, and some medication combinations that may induce toxicity or side effects [29], [30]. Thus, personalized drug-drug interactions (DDIs) [31] analysis is crucial for safeguarding patient health [32], [32].

To address these challenges, numerous deep learning-based medication recommendation methods have been proposed [33], [34], [35], broadly categorized into instance-based and longitudinal models. Instance-based models concentrate on immediate diagnoses for medication advice but overlook past health data [36], [37]. Longitudinal models [38], [39], conversely, adeptly capture the temporal dynamics of a patient's medical record.

Despite the impressive progress of existing models, they still exhibit the following limitations: (i) Inadequate patient-medication interaction learning. Existing methods typically extract patient and medication representations separately using complex networks, followed by employing simple matching functions to predict medication vectors. These approaches often overlook the potential functional link between patient symptoms and medication molecules. (ii) Implicit DDI control. Some methods model DDI using soft constraints, but these often lead to suboptimal performance or uncontrollable DDI.

To address these limitations, we propose a Dual Adaptive Interaction Network (named DAI-Net) and an adaptive balanced DDI strategy for effective and coordinated medication combination recommendation. Our main technical contributions in this work are summarized as follows:

- We propose a dual adaptive interaction network, which integrates medication knowledge and exploits the longitudinal clinical records for effective and coordinated medication combination recommendation.
- We design a symptom-aware medication matching module to extract coordinated associations between patient symptoms and medication molecules, coarse-grained interaction learning. The molecular features are explored to transform symptom-medication matching properties into a symptom-substructure matching matrix, endowing the network with fine-grained interaction information.
- We devise a loss constraint strategy that facilitates gradient optimization to modulate DDI. The model is trained by optimizing the balanced loss between multi-label consistency loss and controllable DDI loss.
- We conduct comprehensive experimental evaluations on the public benchmark MIMIC-III, MIMIC-IV, and eICU datasets compared with state-of-the-art methods. Experimental results demonstrate the superiority of the proposed DAI-Net compared to state-of-the-art methods.

The remainder of this article is organized as follows. In Section II, we review some representative medication combination recommendation methods, including those based on Graph Neural Networks (GNNs), rules, instances, and longitudinal data. Section III provides a detailed formulation of the problem. Section IV introduces the proposed method along with an optimization solution, where dual interaction is depicted in Fig. 2. Extensive experiments and comparisons with state-of-the-art methods are presented in Section V. Finally, Section VI concludes this article with key remarks and conclusions.

## II. Related Works

*Graph neural network for healthcare:* Graph neural networks (GNNs) have been gaining notable traction in healthcare, demonstrating their versatility and efficacy in various applications [40]. In the context of medication correlation, Ma et al. [41]

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utilized GNNs to construct a medication correlation graph, innovatively encoding multi-view medication knowledge. Shang et al. [39], through their work on GAMENet, applied GNNs to encode electronic health records (EHR) and drug-drug interaction (DDI) graphs, thereby integrating essential medication information. Wang et al. [42] developed a dynamic multi-relational medication graph, leveraging GNNs to learn the dependencies among various medication-related relations. SafeDrug, another significant contribution by Yang et al. [34], employed GNNs to enhance medication embeddings based on the molecular structure of medications. Inspired by these approaches, we employ GNNs to encode EHR graphs to enhance medication recommendations.

*Medication combination recommendation:* Medication recommendation has emerged as a promising research area in recent years, driven by the rapid development of medical technology. Existing algorithms for medication combinations fall into three types: rule-based, instance-based, and longitudinal recommendations. Rule-based methods rely on healthcare instructions summarised by the physicians. For example, Gunlicks et al. [43] designed a hard-coded protocol-based recommendation method for habilitative treatment of depression in adolescents. In general, these methods that use fixed rules are poorly generalized. It is also difficult to append traditional rules to a definite knowledge base. Instance-based methods employ the patient's current diagnosis and procedures to model patient representation. For example, Zhang et al. [37] encode current patient visits, modeling within a multi-instance multi-label medication recommendation paradigm. Gong et al. [44] utilize knowledge graphs to discover the relation between patient diseases and medication. However, these approaches neglect to model patients' historical health information. To address this issue, numerous longitudinal-based methods have been proposed to code the longitudinal history of patients. For example, Shang et al. [39] designed memory-based networks with RNN, caching historical medication information as dictionaries to model the longitudinal health information. Yang et al. [34] employed a graph-based MPNN with fingerprints to encode the functionality and connectivity of medication molecules. As well, Yang et al. [33] proposed MICRON to model the longitudinal record changes by a recurrent residual method. Tan et.al [45] designed a set-oriented representation strategy for symptoms and medication.

Despite their initial success, these medication recommendation methods tend to overlook two crucial aspects of patient-medication interaction, which are essential for enhancing accuracy and interpretability. (i) The first overlooked information is the coarse-grained interactions between various types of symptom information in a patient's medical history and the corresponding medication representations, which is crucial for understanding the specific ways that different medication components interact with various symptoms. (ii) The second information is the fine-grained interactions between medication substructures and different types of symptom information. This level of interaction can facilitate the construction of nuanced and disentangled medication representations, thus generating more targeted and effective drug combinations.

To address these gaps, this work proposes a dual adaptive interaction strategy that aims to comprehensively encode the interaction information between patients' multifaceted health records and medication molecules. This approach is designed to significantly enhance the performance of medication recommendation by integrating both coarse-grained and fine-grained interaction information. Enhancing the interpretability of the model by offering clear insights into how different symptoms and medication aspects are linked to each other.

## III. Problem Formulation

*Longitudinal Electrical Health Records:* Longitudinal EHRs encompass the complete medical history of a patient, represented as longitudinal vectors that include diagnoses, procedures, and medications. These vectors can be mathematically expressed as $[x_d^i, x_p^i, y^i]_{i=1}^t$, where $x_d^t \in \{0,1\}^{|\mathcal{D}|}$, $x_p^t \in \{0,1\}^{|\mathcal{P}|}$, $y^t \in \{0,1\}^{|\mathcal{M}|}$, are multi-hot vectors, span across various time points up to the current time point $t$, where $\mathcal{D}, \mathcal{P}, \mathcal{M}$ denote the corresponding sets, respectively.

*Medication Combination Recommendation:* Given a patient's EHR data including diagnosis sequence $[x_d^1, x_d^2, \ldots, x_d^t]$ and procedure sequence $[x_p^1, x_p^2, \ldots, x_p^t]$, medication molecule graph $\mathcal{G}_M$, and DDI adjacency matrix $M_{DDI}$, our objective is to learn a medication combination recommendation function $Rec(\cdot)$ that can produce a multi-label output $\hat{y}^t$, which predicts the appropriate medication combinations at time $t$, where $\hat{y}^t = Rec([x_d^i]_{i=1}^t, [x_p^i]_{i=1}^t, \mathcal{G}_M) \in \{0,1\}^{|\mathcal{M}|}$. During the model training phase, the ground-truth medication combinations ($y^t$) are used as supervisory signals to refine the predictions of $\hat{y}^t$. Additionally, the DDI adjacency matrix ($M_{DDI}$) is utilized to impose controllable DDI constraints on the predicted medication vector $\hat{y}^t$. This pattern ensures that the recommended medications not only follow the patient's medical history but also consider potential DDIs for safety.

## IV. Proposed DAI-Net

In this section, we provide a detailed illustration of the proposed DAI-Net. Fig. 3 shows the overall framework. Our DAI-Net consists of four components: (i) Longitudinal Patient Representation module captures the patient health information from longitudinal clinical data; (ii) Molecular Graph Encoding module outputs medication embedding via a message passing graph neural network. (iii) Symptom-aware Medications Matching module takes the patient representation and medication embedding as inputs and outputs implicit symptom-medication matching matrix. (iv) Fine-grained Medication Representation module harnesses the interaction between symptoms and substructures to modulate medication representation, predicting optimal medication combinations. Further insights can be found in the subsequent sections.

### A. Longitudinal Patient Representation

A patient's health conditions can be encoded by their diagnosis and procedure information, which are crucial to personalized medication recommendations. To fully utilize rich longitudinal

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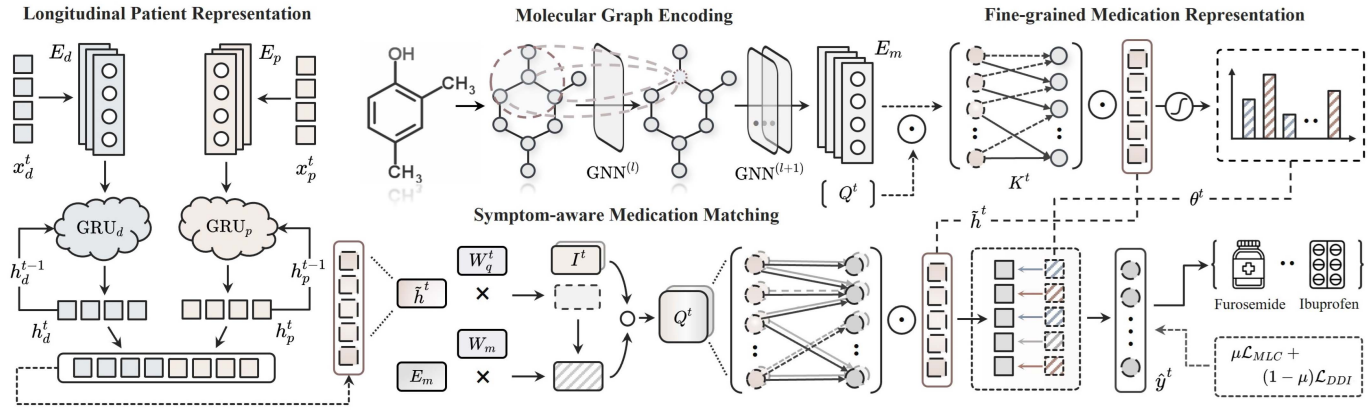



Fig. 3. Overview of our proposed Dual Adaptive Interaction Network (DAI-Net) framework. At the current $t$-th visit, the diagnosis and procedure sequences are input into embedding networks and Dual-GRU to generate the currently hidden states $h_d^t$ and $h_p^t$ from the longitudinal hidden states using (2) and (3). Then, we use a fusion operation to generate a patient health representation. Meanwhile, we obtain medication memory $E_m$ through molecular graph encoding. The symptom-aware medication matching matrix $Q^t$ is obtained through pairwise interaction learning. Finally, we employ a confidence learning strategy to decouple molecular-level dependency for coordinated medication recommendation dynamically.

EHR data, we design two embedding tables, $E_d \in \mathbb{R}^{|\mathcal{D}|\times dim}$ and $E_p \in \mathbb{R}^{|\mathcal{P}|\times dim}$, where each row of $E_d$ or $E_p$ stocks an embedding vector for a particular diagnosis or procedure, and $dim$ denotes the embedding dimension. Given a multi-hot diagnosis vector $x_d^t \in \{0,1\}^{|\mathcal{D}|}$ and a procedure vector $x_p^t \in \{0,1\}^{|\mathcal{P}|}$, we first map the corresponding diagnosis and procedure codes into the embedding space by vector-matrix dot product as follows:

$$d_e^t = x_d^t\, E_d, \quad p_e^t = x_p^t\, E_p, \tag{1}$$

where $d_e^t, p_e^t \in \mathbb{R}^{dim}$, encode the current health condition of the patient. However, a separate visit may be inadequate to determine appropriate therapeutic, e.g., it is problematic to prescribe the same medication to two patients with pneumonia, one who is emergent and another who has long chronic symptoms. Thus, we employ dual recurrent neural networks to track dynamic patient health history, following GAMNet [39] and SafeDrug [34]. Specifically, we use two individual GRU [46] models to capture the hidden temporal dependency information on the diagnosis and procedures,

$$h_d^t = \text{GRU}_d(d_e^t, h_d^{t-1}) = \text{GRU}_d(d_e^1, d_e^2, \ldots, d_e^t), \tag{2}$$

$$h_p^t = \text{GRU}_p(p_e^t, h_p^{t-1}) = \text{GRU}_p(p_e^1, p_e^2, \ldots, p_e^t), \tag{3}$$

where $h_d^t, h_p^t \in \mathbb{R}^{\text{dim}}$, are the hidden embeddings at $t$-visit. Then, we concatenate the diagnosis embedding $h_d^t$ and procedure embedding $h_p^t$ into a compact patient representation $\tilde{h}^t$. Following common practice, we concatenate two vectors and then apply an MLP [47] to obtain the final patient representation $\tilde{h}^t \in \mathbb{R}^{dim}$ as follows:

$$\tilde{h}^t = [\, h_d^t \oplus h_p^t \,]\, W_h, \tag{4}$$

where $W_h \in \mathbb{R}^{2dim\times dim}$ is a learnable parameters matrix.

## B. Molecular Graph Encoding

To capture medication properties and dependencies, we employ a graph neural network to encode the medication molecule graph. In this paper, we use a message-passing neural network (MPNN) [48] with learnable fingerprints in practical implementation. Given a set of all appeared atoms $\mathcal{C} = \{a_i\}$, we create a learnable atom embedding table $E_a \in \mathbb{R}^{|\mathcal{C}|\times dim}$, where each row contains the initial fingerprint for a particular atom. We construct a medication molecule graph $\mathcal{G}_M = \{\mathcal{V}_M, \mathcal{E}_M\}$, atoms $a_1, \ldots, a_n$ as the vertices and atom-atom bonds as edges, $\mathcal{V}_M$, $\mathcal{E}_M$ denote node set and edge set. For the initial atom fingerprints from $E_a$, i.e., $\{m_v^{(0)}\}_{v=1}^n$, we formulate the layer-wise message passing over the graph as follows:

$$m_{\mathcal{N}(v)}^{l+1} = \mathcal{AGG}_{l+1}(m_v^l, m_u^l; W_{vu}), \tag{5}$$

$$m_v^{l+1} = \mathcal{UPD}_{l+1}(m_v^l, m_{\mathcal{N}(v)}^{l+1}), \tag{6}$$

where $m_{\mathcal{N}(v)}^{l+1}$ denotes the encoded messages from the neighbors of atom $v$ at the $l$-th layer, $m_v^{l+1}$ is the hidden state of atom $v$. During message passing, $m_{\mathcal{N}(v)}^{l+1}$ and $m_v^{l+1}$ for each vertex atom $v$ are updated using message function $\mathcal{AGG}_{l+1}(\cdot)$, and vertex update function $\mathcal{UPD}_{l+1}(\cdot)$, and $W_{vu}$ is the layer-wise parameters. Then, the medication molecule global representation is pooled by a Readout function, which calculates the average of all-atom fingerprints as follows:

$$e_m(\mathcal{G}_M) = \mathcal{R}(\{m_v^{(L)}, \forall v \in \mathcal{V}\}). \tag{7}$$

We employ the duplicate MPNN encoder with shared parameters for each single medication molecule, where there are a total of $|\mathcal{M}|$ different medications. We compose all medication molecule embeddings after MPNN into a medication memory $E_m \in \mathbb{R}^{|\mathcal{M}|\times h_m}$, where each row is a medication embedding, $|\mathcal{M}|$ denotes the number of medication set $\mathcal{M}$.

## C. Symptom-Aware Medication Matching

In medication recommendation, we expect to determine the most suitable medication combination for a patient based on their medical history information and current health condition. Conventional medication recommendation approaches directly

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compute matching scores between patient representation and medication memory embedding, without considering that different medication prescriptions may target distinct symptoms. In this work, we propose a symptom-aware medication matching module to capture implicit pairwise interactions between specific symptom types in patients' health records and corresponding medication molecules. Given the hidden patient representation $q^t \in \mathbb{R}^{dim}$ and medication embedding $E_m \in \mathbb{R}^{|\mathcal{M}| \times h_m}$ after GRU and GNN encoders. The pairwise interaction learning can obtain a single-head implicit symptom-medication matching matrix $Q^t \in \mathbb{R}^{dim \times |\mathcal{M}|}$ as follows:

$$Q^t(\tilde{h}^t; E_m) = I^t \odot \left[ (W_q^t \tilde{h}^{t\top}) \cdot (W_m^t E_m^\top) \right], \tag{8}$$

where $W_q^t \in \mathbb{R}^{dim}$, $W_m^t \in \mathbb{R}^{dim \times h_m}$ are learnable weight parameters, $I^t \in \mathbb{R}^{dim \times |\mathcal{M}|}$, and $\odot$ denotes Hadamard product. The elements in $I$ represent the interaction intensity of patient features (can be perceived as symptom profiles) and medication sub-structure pairs, with mapping to potential binding sites. The process can be regarded as an attention mechanism where one medication responds variably to different symptoms. To intuitively understand pairwise interaction, an element $I_{i,j}$ in (8) can also be written as:

$$Q_{i,j}^t = I_{i,j}^t \times \left[ W_q^t \tilde{h}^{t\top} \right]_{i,:} \cdot \left[ W_m^t E_m^\top \right]_{:,j}. \tag{9}$$

As can be seen, we first map patient representation $\tilde{h}^t$ and medication embedding $E_m$ to a joint feature space with $W_q^t$ and $W_m^t$, and we use Hadamard product operation and a learnable interaction intensity matrix $I^t$ to obtain medication implicit matching matrix $h^t$. In this way, the proposed pairwise interaction learning provides interpretability on how symptom-medication pairs contribute to the predicted medication.

### D. Fine-Grained Medication Representation

Although coarse-grained symptom-medication interaction is valuable for the prediction of medication combinations, it is insufficient for mining the molecular-level knowledge of medications and may cause certain performance bounds. To address this dilemma, we utilize symptom-substructure interactions, which represent finer-grained molecular-level information, to obtain disentangled medication representations.

Given the implicit symptom-medication matching matrix $Q^t$, we treat the patient health representation $\tilde{h}^t$ as a query to identify the most suitable medication combination. Moreover, we employ a simple confidence estimate strategy to decouple longitudinal dependency in $Q^t$ at the molecular level. Specifically, we use a transform matrix $W_e$ to obtain symptom-substructure interaction matrix $K^t \in \mathbb{R}^{dim \times h_m}$ as follows:

$$K^t = tanh(Q^t E_m) W_e, \tag{10}$$

where $tanh(\cdot)$ is tanh activation function, $W_e \in \mathbb{R}^{h_m \times h_m}$ is a learnable mapping matrix. We assume that $\tilde{h}$ is a query to calculate medication matching vector and modulate medication representation with medication matching confidence $\theta^t$ to obtain the final predicted medication matching vector:

$$\hat{y}^t = \sigma(\theta^t \circ Q^t \tilde{h}^{t\top}), \tag{11}$$

where $\circ$ denotes point-wise product, $\theta^t = conf\left(K^t \tilde{h}^{t\top}\right) \in \mathbb{R}^{|\mathcal{M}|}$ represents the medication matching confidence scores vector, and $\sigma$ denotes sigmoid activate function. The confidence estimation function $conf(\cdot) : \mathbb{R}^{h_m} \to \mathbb{R}^{|\mathcal{M}|}$ is constructed with a simple fully connected neural network. The final medication vector $\hat{y}^t \in \mathbb{R}^{|\mathcal{M}|}$ consists of the matching scores of $|\mathcal{M}|$ medications to the patient. Further, with a threshold $\delta$, we can recommend a medication combination to the patient, i.e., a multi-hot vector, by picking those entries where the value is greater than $\delta$ as follows:

$$\hat{\mathcal{C}}^t = \left\{ \hat{y}_i^t \mid \hat{y}_i^t > \delta, 1 \le i \le |\mathcal{M}| \right\}. \tag{12}$$

### E. Model Training and Inference

In training, we optimize all the learnable parameters. In the inference phase, the model works in the same pipeline as training. DAI-Net is trained end-to-end via a combined loss.

*Multi-label Consistency Loss:* We formulate medication combination recommendations as a multi-label binary classification task. Let $y^t$ denote the ground-truth multi-hot medication vector, $\hat{y}^t$ indicates the predicted medication recommendation vector. Treating the prediction for each medication as a separate subproblem, we employ *Binary Cross-Entropy* (BCE) loss ($\mathcal{L}_{BCE}$) as part of the objective. Further, we empirically employ the *Multi-Label Hinge* (MLH) loss ($\mathcal{L}_{MLH}$) to ensure that the score of the truth label retains a clear margin over the others to make the results more robust, with the following formula:

$$\mathcal{L}_{MLH} = \frac{1}{|\mathcal{M}|} \sum_{j=1}^{|\mathcal{M}|} \sum_{i=1}^{|\hat{\mathcal{C}}^t|} \max(0, 1 - (\hat{y}^t[\hat{\mathcal{C}}_i^t] - \hat{y}^t[j])), \tag{13}$$

where $\hat{y}[i]$ means the value at $i$-th coordinate at $t$-th visit, $\hat{y}^t[\hat{\mathcal{C}}_i^t]$ means $i$-th predicted label indexed by predicted label set $\hat{\mathcal{C}}^t$ at $t$-th visit. Multi-label Consistency loss is defined as the weighted sum of $\mathcal{L}_{BCE}$ and $\mathcal{L}_{MLH}$:

$$\mathcal{L}_{MLC} = \pi[0]\mathcal{L}_{BCE} + \pi[1]\mathcal{L}_{MLH}, \tag{14}$$

where $\pi[\cdot]$ are pre-defined mixture weights, and meet the required conditions: $\pi[0], \pi[1] \ge 0, \pi[0] + \pi[1] = 1$.

*Controllable DDI Loss:* In the training phase, the accuracy and DDI rate often increase together. However, what we expect is an accurate model with low DDI rates. Based on the DDI adjacency matrix $M_{DDI}$, we design a controllable DDI loss over $\hat{y}^t$ to constrain DDI in the recommendation.

$$\mathcal{L}_{DDI} = \sum_{i=1}^{|\mathcal{M}|} \sum_{j=1}^{|\mathcal{M}|} \left[ M_{DDI} \odot (\hat{y}^{t\top} \hat{y}^t) \right]_{i,j}, \tag{15}$$

where $\hat{y}_t^\top \hat{y}_t \in \mathbb{R}^{|\mathcal{M}| \times |\mathcal{M}|}$ gives the pairwise probabilities of the predicted medications. Intuitively, if two medication representations $i$ and $j$ interact to induce a DDI, then $M_{DDI}[i, j] = 1$. Thus, large pairwise DDI probabilities will lead to large DDI loss values.

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TABLE I
STATISTICS OF THE PROCESSED DATASET

| Items | MIMIC-III | MIMIC-IV | eICU |
|---|---|---|---|
| # of patients | 6350 | 9036 | 3930 |
| # of visits | 15 032 | 20 616 | 27 134 |
| Avg. # of visits | 2.37 | 2.28 | 6.9 |
| Max. # of visits | 29 | 28 | 85 |
| Diag. space size | 1958 | 1892 | 2156 |
| Prod. space size | 1430 | 4939 | - |
| Med. space size | 112 | 131 | 154 |

For coordinated medication recommendations, maintaining a low DDI rate is important. We set a target DDI acceptance rate $\gamma \in (0, 1)$ to constrain the loss. When DDI is under control, we focus on accuracy. But when DDI becomes too high, $\mu$ stays low to prioritize reducing DDI. The overall loss is defined as follows:

$$\mathcal{L}_{Overall} = \mu \mathcal{L}_{MLC} + (1 - \mu)\mathcal{L}_{DDI}, \tag{16}$$

Let the ideal DDI rate be $\gamma$. If the actual DDI rate is greater than the ideal DDI rate, i.e, DDI $\geq \gamma$, we need to optimize $\mathcal{L}_{DDI}$ and allow the model to adaptively adjust the value of $\mu$ depending on the margin. Otherwise, we only optimize the multi-label consistency loss. This dynamic adjustment of $\mu$ allows for an elegant balance between medication coordination and accuracy of recommendation. The formulaic expression of $\mu$ can be written as:

$$\mu = \begin{cases} 1, & \text{DDI} \leq \gamma \\ exp\left(\dfrac{\gamma - \text{DDI}}{\gamma \times K_p}\right), & \text{DDI} > \gamma \end{cases}, \tag{17}$$

where $K_p$ is a correcting factor for the proportional signal.

Different from the linear adjustment function used in [34], $exp(\frac{\gamma-\text{DDI}}{\gamma \times K_p})$ is a concave function. According to the mathematical property of a concave function, when the margin between DDI and $\gamma$ is large, the value of $\mu$ will diminish smoothly during the optimization process so that the model remains optimized in the direction of the DDI loss. When the margin is small, the value of $\mu$ will drop drastically during the iteration and the model will focus on optimizing the multi-label consistency loss. As shown in experiments, $\gamma$ acts as an upper bound on the output DDI rate. By presetting $\gamma$, DAI-Net can meet different target DDI safety levels. This provides a clinically meaningful way to control the DDI-accuracy tradeoff.

The model inference pipeline follows the same process as training. We apply a threshold $\delta = 0.5$ on the output medication representation in (11), selecting medications with values greater than $\delta$ as the final recommendations. The overall algorithm process is outlined in Algorithm 1.

**Algorithm 1:** Pseudocode of the Proposed DAI-Net.

**Require:** Traning set $\mathcal{T}$, hyperparameters $\pi[\cdot]$ in (14), predefined admissible DDI rates $\gamma$, and DDI matrix $M_{DDI}$;
**Ensure:** The proposed DAI-Net model;
Initial the parameters: $E_d$, $E_p$, $E_a$, GRU$_d$, GRU$_p$, $W_h^t$, $I^t$, $W_q^t$, $W_m^t$, $W_e^t$, and $conf\,(\cdot)$;
Generate the medication embedding $E_m$ with (5), (6), (7);
**for** *patient* $i := 1$ to $|\mathcal{T}|$ **do**
  Read the $i$-th patient's history records $[x_d^t]$ and $[x_p^t]$;
  **for** *visit* $t := 1$ to $|[x_d^t]|$ **do**
    Select the $t$-th visit of $i$-th patient, i.e, $x_d^t$, $x_p^t$;
    Generate embeddings $d_e^t$ and $p_e^t$ by (1);
    Generate hidden embeddings $h_d^t$ and $h_p^t$ by (2), (3);
    Obtain patient health representation $\tilde{h}^t$ by (4);
    Obtain medication matching matrix $Q^t$ with (8);
    Obtain interaction matrix $K^t$ in (10);
    Obtain medication representation with (11);
    Generate medication combination via (12);
  **end for**
  Calculate $\mathcal{L}_{MLC}$, $\mathcal{L}_{DDI}$ in (14), (15), and accumulate;
**end for**
Choose $\gamma$ and Optimizing the overall loss in (16);

## V. EXPERIMENTS

In this section, we evaluate our proposed DAI-Net framework focusing on the following four research questions:

- RQ1: How does DAI-Net perform compared to state-of-the-art drug or medication recommendation methods?
- RQ2: What are the effects of different components?
- RQ3: How do the hyper-parameters affect performance?
- RQ4: What real medications are recommended by DAI-Net and how are they coordinated?

### A. Experimental Settings

*Dataset:* To demonstrate the effectiveness of our proposed DAI-Net model, we conducted experiments under publicly available EHR datasets, i.e., MIMIC-III [49], MIMIC-IV [50] and eICU [51] datasets. Detailed statistics of processed data are shown in Table I. The diagnosis and procedure sequences are coded by ICD-9[1]. We extract the DDI of the Top-40 severity types from Twosides database [52], where DDI data are reported by ATC Third Level codes. eICU dataset only contains the patient's diagnosis and medication records.

*Evaluation Protocols:* We use four common metrics, including Jaccard similarity score (Jaccard), F1 score, Precision-Recall Area Under Curve (PRAUC), and DDI rate, where DDI rate is a safety measurement that computes the rate of paired medications response among predicted medications. We provide a detailed definition of evaluation metrics:

Jaccard coefficient. We take the average overall visits to calculate the final Jaccard similarity score for patient-$x$ as,

$$\text{Jaccard} = \frac{1}{N_x}\sum_{t=1}^{N_x}\frac{|\{i : y_i^t = 1\} \cap \{i : \hat{y}_i^t = 1\}|}{|\{i : y_i^t = 1\} \cup \{i : \hat{y}_i^t = 1\}|}, \tag{18}$$

where $\cap$ is set interaction and $\cup$ is set union operation, $N_x$ denotes the total number of visits for patient-$x$.

[1]http://www.icd9data.com/

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F1 score. The formula is the harmonic mean of precision and recall. For patient-$x$, F1 score is calculated as:

$$\text{F1-score} = \frac{2}{N_x}\sum_{t=1}^{N_x}\frac{|\{i : y_i^t = 1\} \cap \{i : \hat{y}_i^t = 1\}|}{|\{i : \hat{y}_i^t = 1\}| + |\{i : y_i^t = 1\}|}. \quad (19)$$

PRAUC. We treat the medication recommendation as a retrieval problem. The PRAUC is calculated as follows:

$$\text{PRAUC} = \frac{1}{N_x}\sum_{t=1}^{N_x}\sum_{k=1}^{|\mathcal{M}|}\text{Precision}(k)^t\Delta\text{Recall}(k)^t, \quad (20)$$

$$\text{Precision}^t = \frac{|\{i : y_i^t = 1\} \cap \{i : \hat{y}_i^t = 1\}|}{|\{i : \hat{y}_i^t = 1\}|}, \quad (21)$$

$$\Delta\text{Recall}(k)^t = \text{Recall}(k)^t - \text{Recall}(k-1)^t, \quad (22)$$

$$\text{Recall}^t = \frac{|\{i : y_i^t = 1\} \cap \{i : \hat{y}_i^t = 1\}|}{|\{i : y_i^t = 1\}|}, \quad (23)$$

where $k$ is the rank in the sequence of the retrieved medications, $\text{Precision}(k)^t$ is the precision at cut-off $k$ in the ordered retrieval list, and $\Delta\text{Recall}(k)^t$ is the change in recall from medication $k-1$ to $k$.

DDI Rate. The DDI rate is defined as the number of DDI pairs present in the predicted medication divided by the number of pairs of possible medication combinations as:

$$\text{DDI Rate} = \frac{1}{N_x}\sum_{t=1}^{N_x}\frac{\sum_{l,k=1}^{|\hat{\mathcal{C}}^t|}\mathbf{1}\{[M_{DDI}]_{lk} = 1\}}{\sum_{l,k=1}^{|\hat{\mathcal{C}}^t|}1}, \quad (24)$$

where $\hat{\mathcal{C}}^t$ is the predicted result, and $\mathbf{1}\{\cdot\}$ is a function that return 1 when expression in $\{\cdot\}$ is true, otherwise return 0.

*Compared methods:* We compare DAI-Net with the state-of-the-art methods as follows: standard Logistic Regression (LR), Ensemble Classifier Chain (ECC) [53] is a multi-label classification approach, RETAIN [54] is an RNN-based model, LEAP [37] is an instance-base approach, DMNC [38] is a longitudinal memory-based method, ManyDG [55] introduces a domain generalization strategy that treats each patient as a separate domain. While GAMENet [39], CompNet [42], SafeDrug [34], 4SDrug [45], and DPG [56] utilize additional medical knowledge, e.g., DDI knowledge, to ensure safety.

*Implementation Details:* Our method is implemented by PyTorch 1.11.0 based on Python 3.7.16 and tested on an Intel Xeon E5-2678 machine with 256 GB RAM and an Nvidia Titan V GPU. We choose the optimal hyper-parameters based on the validation set. Models are trained on Adam optimizer with $1 \times 10^{-3}$ learning rate and batch size 16 for 50 epochs. We fixed the random seed as 1203 for PyTorch to ensure the reproducibility of the models. We set correcting factor $\kappa$ is 0.05 and $\gamma$ is 0.07. The weight $\pi[0]$ in our $\mathcal{L}_{MLC}$ is 0.95. Precisely, we random sample 80% data from the test set for a round of evaluation, and the results of 10 rounds are used to calculate the mean and standard deviation, which are finally reported. As the sampling process is random, the final experimental results may vary slightly.

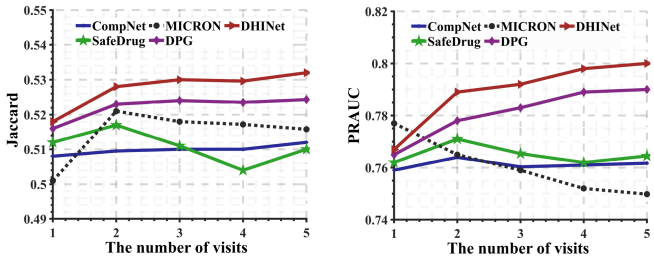



Fig. 4. The line graphs illustrate the effect of the number of visits.

## B. Experimental Results (RQ1)

The experimental results of different methods are shown in Tables II and IV. For traditional classification methods that are not deep learning, e.g., LR and ECC models recommend fewer medications overall but have high DDI rates in their predicted combinations. As can be seen, the ECC, DMNC, and LEAP models that treat recommendation as a sequence generation task show poor performance. Formulating recommendations as multi-label classification appears more effective. Thus, we formulate the medication recommendation task as multi-label classification might be more straightforward and reasonable. LR and RETAIN do not consider DDI knowledge, producing excessive interactions. GAMENet bases predictions on historical data, but real records contain high baseline DDI rates, limiting safety. SafeDrug capitalizes on DDI knowledge while preserving effectiveness and safety. 4SDrug, ManyDG, and DPG do not fully mine patient-drug interaction information. From Tables II and IV, we can see that the performance of DAI-Net outperforms the other baselines.

The model complexity of DAI-Net and other DL-based compared methods is assessed in Table III. Compared to LEAP, DMNC, GAMENet, and CompNet, our model has relatively low spatial and temporal complexity. We have reason to agree that DAI-Net is efficient and flexible. In order to investigate the capability of the DAI-Net to model patient medical information during historical visits, we conducted a comparative experiment to probe the impact of the number of visits in the MIMIC-III, and MIMIC-IV datasets. The experimental results are shown in Fig. 4, which demonstrate that DAI-Net almost achieves the best performance under different visit periods. It can be seen that the performance of both DAI-Net and DPG has progressively improved with an increase in the number of visits.

## C. Ablation Study (RQ2)

To better understand the contributions of the components and configurations of the proposed DAI-Net, we conduct comprehensive ablation studies on MIMIC-III and MIMIC-IV datasets. Specifically, we design the following variants: Remove diagnosis information in visits (called DAI-Net-$\mathcal{D}$); Drop procedure information in visits (called DAI-Net-$\mathcal{P}$); Delete the symptom-aware medication matching module and use dot product between the patient representation and medication embedding to obtain recommended medication (called DAI-Net-$Q$); Remove

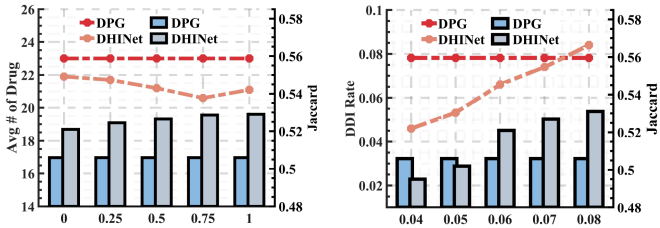
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TABLE II
PERFORMANCE COMPARISON OF DIFFERENT METHODS ON MIMIC-III DATASET. THE GROUND-TRUTH DDI RATE IS 0.0810

| Model | Jaccard | F1-score | PRAUC | Avg # of Drug | DDI Rate |
|---|---|---|---|---|---|
| LR | $0.4865 \pm 0.0021$ | $0.6434 \pm 0.0019$ | $0.7509 \pm 0.0018$ | $16.1773 \pm 0.0942$ | $0.0829 \pm 0.0009$ |
| ECC [56] | $0.4856 \pm 0.0035$ | $0.6438 \pm 0.0031$ | $0.7590 \pm 0.0029$ | $16.9709 \pm 0.1812$ | $0.0817 \pm 0.0015$ |
| RETAIN [57] | $0.4887 \pm 0.0028$ | $0.6481 \pm 0.0027$ | $0.7556 \pm 0.0033$ | $20.4925 \pm 0.2832$ | $0.0835 \pm 0.0020$ |
| LEAP [40] | $0.4521 \pm 0.0024$ | $0.6138 \pm 0.0026$ | $0.6467 \pm 0.0034$ | $18.7138 \pm 0.0667$ | $0.0731 \pm 0.0008$ |
| DMNC [41] | $0.4864 \pm 0.0025$ | $0.6529 \pm 0.0030$ | $0.7580 \pm 0.0039$ | $20.0000 \pm 0.0000$ | $0.0842 \pm 0.0011$ |
| GAMENet [42] | $0.5037 \pm 0.0018$ | $0.6601 \pm 0.0019$ | $0.7653 \pm 0.0026$ | $27.2603 \pm 0.1533$ | $0.0859 \pm 0.0006$ |
| CompNet [45] | $0.5051 \pm 0.0032$ | $0.6625 \pm 0.0030$ | $0.7642 \pm 0.0027$ | $23.1274 \pm 0.2136$ | $0.0773 \pm 0.0007$ |
| MICRON [36] | $0.5097 \pm 0.0037$ | $0.6652 \pm 0.0035$ | $0.7687 \pm 0.0028$ | $18.1361 \pm 0.2172$ | $0.0703 \pm 0.0007$ |
| SafeDrug [37] | $0.5117 \pm 0.0035$ | $0.6621 \pm 0.0031$ | $0.7606 \pm 0.0029$ | $19.0607 \pm 0.1835$ | $0.0623 \pm 0.0006$ |
| 4SDrug [48] | $0.4837 \pm 0.0023$ | $0.6456 \pm 0.0026$ | $0.7625 \pm 0.0020$ | $16.2162 \pm 0.0782$ | $0.0693 \pm 0.0009$ |
| ManyDG [58] | $0.4156 \pm 0.0051$ | $0.5787 \pm 0.0049$ | $0.6921 \pm 0.0057$ | $15.5162 \pm 0.1663$ | $0.0796 \pm 0.0012$ |
| DPG [59] | $0.5028 \pm 0.0033$ | $0.6632 \pm 0.0032$ | $0.7648 \pm 0.0022$ | $21.3502 \pm 0.1392$ | $0.0739 \pm 0.0008$ |
| DAI-Net$_{\ominus}$ | $0.5279 \pm 0.0022$ | $0.6794 \pm 0.0026$ | $0.7702 \pm 0.0025$ | $22.5231 \pm 0.1406$ | $0.0795 \pm 0.0007$ |
| DAI-Net | $0.5292 \pm 0.0024$ | $0.6834 \pm 0.0027$ | $0.7738 \pm 0.0023$ | $22.0831 \pm 0.1382$ | $0.0782 \pm 0.0009$ |

TABLE III
DL-BASED MODEL COMPLEXITY COMPARISON

| Model | # of Param. | Training(s) | Inference(s) |
|---|---|---|---|
| RETAIN [57] | 287 940 | 43.72 / Epoch | 4.39 |
| LEAP [40] | 433 286 | 380.17 / Epoch | 43.28 |
| DMNC [41] | 525 533 | 4921.3 / Epoch | 280.25 |
| GAMENet [42] | 449 092 | 183.86 / Epoch | 37.53 |
| CompNet [45] | 364 358 | 167.47 / Epoch | 19.94 |
| SafeDrug [37] | 300 262 | 236.35 / Epoch | 16.37 |
| 4SDrug [48] | 80 348 | 25.76 / Epoch | 15.62 |
| ManyDG [58] | 258 506 | 120.29 / Epoch | 23.75 |
| DPG [59] | 347 595 | 286.87 / Epoch | 21.36 |
| DAI-Net | 362 947 | 350.35 / Epoch | 18.13 |




(a) Varying $\pi[0]$ (b) Varying $\gamma$

Fig. 5. Performance for varying the weight of multi-label consistency loss (i.e., $\pi[0]$) and the expected DDI rate (i.e, hyper-parameter $\gamma$).

fine-grained symptom-substructure interaction learning (called DAI-Net-$K$); Remove controllable DDI loss and only optimize multi-label consistency loss (called DAI-Net-$\gamma$); Change our DDI constraint strategy to the linear adjustment in [34] (called DAI-Net$_{\ominus}$, experimental results are shown in Tables II and IV).

Table V presents the results of the ablation study for the different variants of DAI-Net. As expected, not including diagnosis or procedure information caused a significant decline in model performance, highlighting the importance of utilizing multi-source knowledge. Furthermore, excluding the symptom-aware drug matching module caused decreased performance, emphasizing the importance of patient-drug interaction learning. The ablation of fine-grained symptom-substructure interaction learning also caused harm, which shows the effectiveness of the fine-grained drug representation module. The full DAI-Net outperforms all variants versions (except DAI-Net-$\gamma$ which ignores DDI constraints), affirming the indispensable modeling capabilities offered by each proposed component. In summary, all observations indicate that DAI-Net can effectively utilize longitudinal patient visit information and DDI knowledge to generate coordinated medication combination recommendations for patient treatment.

### D. Hyperparameter Study (RQ3)

Our proposed DAI-Net framework introduces two key hyperparameters: $\pi[0]$ controls the weight of $\mathcal{L}_{BCE}$ and $\mathcal{L}_{MLH}$, $\gamma$ controls the DDI bound to balance safety and accuracy. In this study, we delineate the impact of these two hyperparameters on performance outcomes, while concurrently elucidating the optimal strategies for their configuration.

We show the performance for varying the weight of $\mathcal{L}_{BCE}$ and $\mathcal{L}_{MLH}$ (i.e., $\pi[0]$) and the hyperparameter $\gamma$ on MIMIC-III. in Fig. 5. $\mathcal{L}_{BCE}$ controlled by $\pi[0]$ can control the number of medications, and it can be seen that Avg # of output medications decreases as $\pi[0]$ increases. However, too large $\pi[0]$ may lead to overfitting of the model. We found the optimal $\pi[0]$ on MIMIC-III with a value of about 0.95. $\pi[0]$ in the range [0, 1] can be obtained by fine-tuning.

Besides that, the performance varying with $\gamma$ is shown in Fig. 5. Our analysis of the DDI controllability via $\mathcal{L}_{DDI}$ reveals that $\gamma$ effectively balances safety and accuracy. Specifically, $\gamma$ regulates the DDI of the predicted medication combinations, where smaller values of $\gamma$ enhance accuracy, allowing the model to be trained with $\mathcal{L}_{MLC}$ to achieve higher precision. However, increasing $\gamma$ beyond the optimal value results in a decline in accuracy while simultaneously increasing DDI. Empirically, $\gamma = 0.07$ appears to be the optimal choice, serving as a practical guideline.

To investigate the effect of the embedding dimension d on model performance, we conducted experiments varying d from 8 to 512. As illustrated in Fig. 6, the prediction performance improved as d increased initially, but began to plateau or even decline as d continued to grow. These results demonstrate that an appropriate embedding dimension can enhance performance,

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TABLE IV
PERFORMANCE COMPARISON OF DIFFERENT METHODS ON MIMIC-IV AND EICU DATASETS

| Model | MIMIC-IV | | | eICU | | |
|---|---|---|---|---|---|---|
| | Jaccard | F1-score | PRAUC | Jaccard | F1-score | PRAUC |
| LR | $0.4732 \pm 0.0018$ | $0.6303 \pm 0.0017$ | $0.7380 \pm 0.0016$ | $0.4026 \pm 0.0023$ | $0.5548 \pm 0.0022$ | $0.7364 \pm 0.0027$ |
| ECC | $0.4403 \pm 0.0010$ | $0.6016 \pm 0.0009$ | $0.7139 \pm 0.0006$ | $0.3289 \pm 0.0018$ | $0.4634 \pm 0.0013$ | $0.7334 \pm 0.0015$ |
| RETAIN | $0.4398 \pm 0.0014$ | $0.6013 \pm 0.0008$ | $0.7140 \pm 0.0002$ | $0.4118 \pm 0.0021$ | $0.5590 \pm 0.0018$ | $0.6992 \pm 0.0019$ |
| Leap | $0.4350 \pm 0.0023$ | $0.5963 \pm 0.0020$ | $0.6321 \pm 0.0040$ | $0.3033 \pm 0.0027$ | $0.4358 \pm 0.0024$ | $0.4826 \pm 0.0023$ |
| DMNC | $0.4673 \pm 0.0025$ | $0.6322 \pm 0.0022$ | $0.7389 \pm 0.0034$ | $0.4259 \pm 0.0026$ | $0.5784 \pm 0.0023$ | $0.7235 \pm 0.0028$ |
| GAMENet | $0.4784 \pm 0.0017$ | $0.6364 \pm 0.0017$ | $0.7395 \pm 0.0028$ | $0.4519 \pm 0.0019$ | $0.6008 \pm 0.0017$ | $0.7371 \pm 0.0025$ |
| CompNet | $0.4816 \pm 0.0028$ | $0.6407 \pm 0.0023$ | $0.7410 \pm 0.0019$ | $0.4563 \pm 0.0029$ | $0.6122 \pm 0.0023$ | $0.7394 \pm 0.0027$ |
| MICRON | $0.4382 \pm 0.0025$ | $0.6032 \pm 0.0026$ | $0.7106 \pm 0.0018$ | $0.3703 \pm 0.0026$ | $0.5174 \pm 0.0021$ | $0.7346 \pm 0.0019$ |
| SafeDrug | $0.4837 \pm 0.0062$ | $0.6435 \pm 0.0054$ | $0.7401 \pm 0.0031$ | $0.4422 \pm 0.0037$ | $0.5903 \pm 0.0032$ | $0.7378 \pm 0.0035$ |
| 4SDrug | $0.4792 \pm 0.0022$ | $0.6368 \pm 0.0019$ | $0.7353 \pm 0.0017$ | $0.4226 \pm 0.0033$ | $0.5856 \pm 0.0026$ | $0.7207 \pm 0.0024$ |
| ManyDG | $0.4736 \pm 0.0036$ | $0.6325 \pm 0.0028$ | $0.7347 \pm 0.0026$ | $0.4077 \pm 0.0028$ | $0.5603 \pm 0.0027$ | $0.7042 \pm 0.0026$ |
| DPG | $0.4795 \pm 0.0027$ | $0.6407 \pm 0.0025$ | $0.7383 \pm 0.0023$ | $0.4534 \pm 0.0025$ | $0.5893 \pm 0.0021$ | $0.7265 \pm 0.0022$ |
| DAI-Net$_{\ominus}$ | $0.4956 \pm 0.0018$ | $0.6518 \pm 0.0015$ | $0.7424 \pm 0.0024$ | $0.4665 \pm 0.0022$ | $0.6097 \pm 0.0018$ | $0.7388 \pm 0.0020$ |
| DAI-Net | $0.5002 \pm 0.0016$ | $0.6558 \pm 0.0012$ | $0.7457 \pm 0.0029$ | $0.4697 \pm 0.0021$ | $0.6126 \pm 0.0019$ | $0.7432 \pm 0.0018$ |

TABLE V
ABLATION STUDY FOR DIFFERENT COMPONENTS OF DAI-NET ON MIMIC-III AND MIMIC-IV DATASETS

| Model | MIMIC-III | | | MIMIC-IV | | |
|---|---|---|---|---|---|---|
| | Jaccard | F1-score | PRAUC | Jaccard | F1-score | PRAUC |
| DAI-Net-$\mathcal{D}$ | $0.4826 \pm 0.0030$ | $0.6421 \pm 0.0024$ | $0.7567 \pm 0.0026$ | $0.4795 \pm 0.0022$ | $0.6418 \pm 0.0018$ | $0.7398 \pm 0.0026$ |
| DAI-Net-$\mathcal{P}$ | $0.4879 \pm 0.0026$ | $0.6433 \pm 0.0025$ | $0.7434 \pm 0.0030$ | $0.4802 \pm 0.0022$ | $0.6427 \pm 0.0019$ | $0.7405 \pm 0.0025$ |
| DAI-Net-$Q$ | $0.4983 \pm 0.0038$ | $0.6537 \pm 0.0034$ | $0.7583 \pm 0.0028$ | $0.4768 \pm 0.0028$ | $0.6349 \pm 0.0023$ | $0.7399 \pm 0.0027$ |
| DAI-Net-$K$ | $0.5163 \pm 0.0032$ | $0.6724 \pm 0.0028$ | $0.7652 \pm 0.0029$ | $0.4927 \pm 0.0027$ | $0.6485 \pm 0.0020$ | $0.7420 \pm 0.0033$ |
| DAI-Net-$\gamma$ | $0.5312 \pm 0.0027$ | $0.6872 \pm 0.0029$ | $0.7793 \pm 0.0026$ | $0.5089 \pm 0.0021$ | $0.6613 \pm 0.0018$ | $0.7496 \pm 0.0024$ |
| DAI-Net | $0.5292 \pm 0.0024$ | $0.6834 \pm 0.0027$ | $0.7738 \pm 0.0023$ | $0.5002 \pm 0.0016$ | $0.6558 \pm 0.0012$ | $0.7457 \pm 0.0029$ |

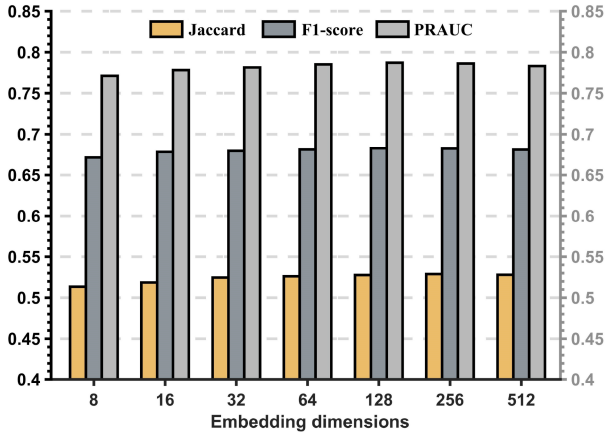



Fig. 6. Effect of embedding dimension d on different evaluation metrics.

while an excessively large d risks overfitting. An embedding size in the range of 64 to 256 provided optimal performance, balancing representational capacity and generalization. Further, increasing model capacity through larger embeddings yielded diminishing returns and reduced generalization capability. In summary, our experiments delineate the influence of embedding dimensionality on performance, and identify the embedding sizes that maximized predictive accuracy while avoiding overfitting.

## E. Case Study (RQ4)

To demonstrate the advantages of DAI-Net over the three medication recommendation baselines, we demonstrate the recommended medications predicted by SafeDrug, 4SDrug, DPG, and the proposed DAI-Net on the MIMIC-III dataset.

We analyze a complex patient case with two visits from the MIMIC-III test set to compare methods. Table VI shows the prescribed and predicted medications. For (case 1) 1st visit, DAI-Net achieved the best performance with 16 correct medications, 4 unseen medications, and 1 missed medication. SafeDrug predicts 15 correct medications and 5 unseen medications, and these predicted medications have relatively few DDIs. 4SDrug predicts 14 correct medications, 3 unseen medications, and 3 medications missed, which indicates that this approach tends to predict fewer medications, resulting in a large number of missing medications. DPG predicts 15 correct medications, 6 unseen medications, and 2 missed medications. Although DPG predicted 15 correct medications, the medication combination with a high DDI would affect safety. DAI-Net predicts a total of 20 medicines for this patient and the number of correct medicines is 17. We can see that DAI-Net is more effective in predicting coordinated and trusted medication combinations but also in the number of medications. For (case 1) 2nd visit, the situation is analogous. DAI-Net achieved the best performance with 17 correct medications, 4 unseen medications, and 1 medication missed.

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TABLE VI
EXAMPLE RECOMMENDED MEDICATIONS FOR TWO CASES ON MIMIC-III. HERE "CORRECT" INDICATES CORRECT PREDICTION, "UNSEEN" REFERS TO PREDICTED MEDICATIONS NOT IN THE GROUND TRUTH, WHILE "MISSED" REFERS TO MEDICATIONS THAT WERE IN THE GROUND TRUTH BUT NOT PREDICTED. THE MEDICATION PAIRS DENOTED BY THE SAME NON-BLACK COLORS HAVE HARMFUL DDIS. BEST VIEWED IN COLOR

| Diagnosis | Methods | Recommended Medication Combination |
|---|---|---|
| (Case 1) 1st visit<br>Malignant neoplasm of brain<br>Hyperlipidmia<br>Gout | Ground-truth (Num: 17) | B05C, A07A, C07A, A06A, N02B, C02D, B01A, A02B, A12C, H03A, N03A, A01A, N05A, C09A, C10A, C08C, A12B |
| | SafeDrug (Num: 20) | 15 correct + 5 unseen (N06A, C10A, J01D, A04A, H01C) + 2 missed |
| | 4SDrug (Num: 17) | 14 correct + 3 unseen (N05C, A04A, N05B) + 3 missed |
| | DPG (Num: 20) | 15 correct + 5 unseen (A07A, N02A, B01A, N05C, C09A) + 2 missed |
| | DAI-Net (Num: 20) | 16 correct + 4 unseen (N02A, N06A, J01D, A04A) + 1 missed |
| (Case 1) 2nd visit<br>Malignant neoplasm of brain Cerebral Edema Hyperchol-esterolemia Gout | Ground-truth (Num: 18) | N02B, A01A, A02B, A06A, A12A, B05C, A12C, C01C, A10A, N01A, C07A, C03C, A12B, N07A, C02D, C01B, J01D, C10A |
| | SafeDrug (Num: 21) | 16 correct + 5 unseen (A07A, A11C, N02A, N02A, C01D) + 2 missed |
| | 4SDrug (Num: 17) | 14 correct + 3 unseen (A07A, N02A, A02A) + 4 missed |
| | DPG (Num: 20) | 15 correct + 5 unseen (A07A, N02A, B01A, N05C, C01D) + 3 missed |
| | DAI-Net (Num: 21) | 17 correct + 4 unseen (A07A, N02A, N02A, C01D) + 1 missed |
| (Case 2) 1st visit<br>Cardiovascular Arrhythmias<br>AV block | Ground-truth (Num: 14) | A02B, A06A, A12C, C07A, C02D, N02A, N06A, C02A, B01A, N03A, C08C, A04A, A07E, N05B |
| | SafeDrug (Num: 18) | 13 correct + 5 unseen (A01A, A12C, B05C, C09A, V03A) + 1 missed |
| | 4SDrug (Num: 15) | 11 correct + 4 unseen (A01A, A12C, B05C, N05A) + 3 missed |
| | DPG (Num: 17) | 12 correct + 5 unseen (N02B, A12C, B05C, C09A, V03A) + 2 missed |
| | DAI-Net (Num: 20) | 14 correct + 6 unseen (N02B, A01A, B05C, N05A, D04A, V03A) + 0 missed |
| (Case 2) 2nd visit<br>Cardiovascular Ventricular disorders<br>Cardiomyopathy | Ground-truth (Num: 13) | A02B, A06A, B05C, C07A, C03C, C02D, N02A, N06A, C02A, B01A, C08C, A04A, N05B |
| | SafeDrug (Num: 17) | 12 correct + 5 unseen (N02B, A12C, C09A, N05A, V03A) + 1 missed |
| | 4SDrug (Num: 15) | 11 correct + 4 unseen (N02B, A01A, C09A, V03A) + 4 missed |
| | DPG (Num: 17) | 11 correct + 6 unseen (N02B, A01A, A12C, C09A, D04A, V03A) + 2 missed |
| | DAI-Net (Num: 18) | 12 correct + 6 unseen (N02B, A12C, C09A, N03A, D04A, V03A) + 1 missed |

Combined with the previous experimental results, the case study further validates the effectiveness of our DAI-Net in terms of its accuracy and DDI control capability.

## VI. CONCLUSION

In this paper, we propose a novel Dual Adaptive Interaction Network (DAI-Net) for coordinated medication recommendation, which can provide prescribing assistance. Specifically, we design a symptom-aware medication matching module to capture implicit patient-medication interaction, encoding coordinated associations between medication molecules and patient symptoms. The molecular features are explored to transform patient-medication interactions into a symptom-substructure interaction matrix for medication representation, endowing the network with fine-grained interaction. Extensive experiments on the MIMIC-III, MIMIC-IV, and eICU datasets demonstrate that the proposed DAI-Net achieves competitive performance compared to the state-of-the-art methods.

## AUTHOR CONTRIBUTIONS

Conceptualization, Xin Zou; methodology, Xin Zou; formal analysis and investigation, Chang Tang, Xiao He, and Zheng Xiao; resources and data curation, Wei Zhang and Jiajia Chen; writing–original draft preparation, Xin Zou; writing–review and editing, Xin Zou and Chang Tang. All authors have read and agreed to the published version of the manuscript.

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